PLANNER RECOVERABILITY SCORES UNDER PHYSICAL DEVIATION: A NEGATIVE ICLR-MAIN EVIDENCE AUDIT

Anonymous authors
Paper under double-blind review

ABSTRACT

Robotic task-and-motion planners usually score plans by nominal feasibility, expected cost, or static risk. The research bet tested here is that a plan should instead be scored by *recoverability*: when execution deviates because of contact, slip, narrow passages, or hidden irreversible traps, can the robot detect the deviation and return to a goal-achieving branch without excessive cost or damage? We implemented a deterministic, reproducible local benchmark with five held-out physical-shift regimes, eight planners, seven random seeds, 11,760 main rollouts, 2,058 ablation rollouts, and 25,200 stress-rollout evaluations. The resulting evidence is useful but not submission-ready for ICLR main. On the combined hard-shift split, the proposed recoverability-score planner reaches 0.850 goal success, but a learned failure classifier reaches 0.878 and an oracle reaches 0.874. The proposed method only improves over a contingent replanner by 0.0068 paired goal-success points, while also causing more irreversible failures and more damage. Worse, removing the irreversible-trap penalty or failure-probability term improves success. We therefore mark this paper **KILL/ARCHIVE**, not because the idea is empty, but because the present evidence falsifies the stronger claim and lacks hardware or high-fidelity simulator validation. Submission-hardening version: v4.

1 TERMINAL DECISION

Decision: KILL/ARCHIVE for ICLR main. The repository now contains a real executable benchmark, baselines, ablations, stress sweeps, figures, and reproducibility files. However, the empirical story does not support a main-conference robotics submission. The proposed recoverability score is not the best non-oracle method on the hardest split, its advantage over a contingent replanner is statistically small, and its own ablations expose a misspecified objective. The result should be archived as a useful negative study rather than submitted as a positive method paper.

2 QUESTION

The intended claim was:

> Score robot plans by recoverability under physical deviation, not only by nominal feasibility.

That claim is plausible in robotics settings where a short plan can enter an unrecoverable contact state, lose an object, or consume all slack before detecting a deviation. A publishable version would need evidence that a recoverability score changes plan choice in a way that consistently beats strong replanning and learned failure predictors under physically meaningful shifts.

3 BENCHMARK

We built a local task-and-motion recoverability benchmark with the following structure.

Table 1: Combined hard-shift results over seven seeds. Confidence intervals are 95% intervals over seed means.

Method	Goal success	Recovery success	Irreversible	Damage
Nominal shortest	0.588 ± 0.033	0.049	0.170	0.265
Risk-averse	0.653 ± 0.056	0.612	0.007	0.031
Robust margin	0.340 ± 0.047	0.576	0.007	0.048
Expected-cost replanner	0.840 ± 0.049	0.602	0.017	0.048
Contingent replanner	0.844 ± 0.041	0.829	0.000	0.010
Learned failure classifier	$\mathbf{0.878 \pm 0.041}$	**0.901**	0.000	**0.007**
Recoverability score	0.850 ± 0.042	0.744	0.007	0.017
Oracle upper bound	0.874 ± 0.040	0.872	0.000	0.010

- **Episodes.** Each episode contains a hidden deviation profile, a budget, a slip factor, a narrow-passage difficulty, a trap severity, and sensor uncertainty.
- **Candidate plans.** The planner chooses among direct shortcuts, robust-margin routes, contingency branches, diagnostic staging, and conservative backoff plans.
- **Splits.** Evaluation covers open easy tasks, narrow-passage shifts, object-slip shifts, irreversible-commitment tasks, and a combined hard-shift split.
- **Rollout coupling.** For fairness, the same episode-plan pair has the same stochastic physical outcome regardless of which method selects it.
- **Metrics.** We report goal success, recovery success after deviation, irreversible failure, damage, total cost, budget overrun, and calibration error across seven seeds.

The benchmark is deterministic from source code and writes all CSV files under `results/`. It is not a real robot benchmark and not a high-fidelity simulator. That limitation is central to the terminal decision.

4 METHODS

We compare the proposed score against seven alternatives:

- **Nominal shortest plan:** chooses the lowest nominal cost.
- **Risk-averse plan:** avoids predicted failure probability.
- **Expected-cost replanner:** optimizes expected physical cost.
- **Robust-margin planner:** prefers geometric slack.
- **Contingent replanner:** values contingency branches and fallback feasibility.
- **Learned failure classifier:** a ridge classifier trained on generated training episodes.
- **Oracle recoverability upper bound:** uses hidden rollout probabilities as a ceiling.
- **Recoverability-score planner:** combines success probability, recovery success, recovery cost, diagnostic value, and irreversible-trap penalties.

5 MAIN RESULTS

Table 1 shows the combined hard-shift split. The proposed score is much better than nominal, risk-only, and robust-margin planning. That is the useful part of the archive. But it does not beat the learned failure classifier and it barely beats contingent replanning.

6 ABLATION RESULTS

The ablation table is the strongest reason to archive the submission. A well-specified recoverability objective should degrade when central terms are removed. Instead, removing the irreversible-trap penalty and removing the failure-probability term both improve hard-split success.

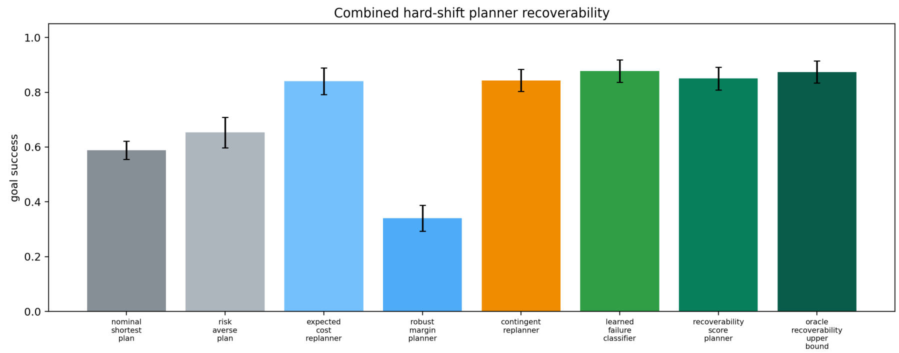

Figure 1: Goal success across evaluation splits. The proposed score is useful on hard splits but does not dominate the learned failure classifier.

Table 2: Recoverability-score ablations on the combined hard-shift split.

Ablation	Goal success	Recovery success	Irreversible	Damage
Full score	0.850 ± 0.042	0.744	0.007	0.017
Minus recovery success	0.844 ± 0.045	0.655	0.007	0.027
Minus recovery cost	0.847 ± 0.038	0.723	0.007	0.020
Minus irreversible-trap penalty	$\mathbf{0.867 \pm 0.041}$	$\mathbf{0.886}$	$\mathbf{0.000}$	$\mathbf{0.010}$
Minus diagnostic value	0.837 ± 0.046	0.593	0.007	0.034
Minus failure probability	0.864 ± 0.033	0.833	0.003	0.014
Risk-only score	0.687 ± 0.048	0.588	0.007	0.031

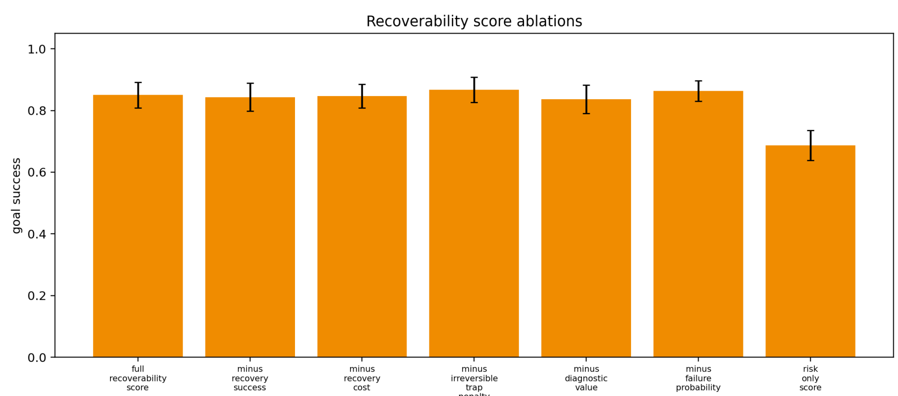

Figure 2: Ablations reveal that two removals improve the proposed method, which blocks a positive method claim.

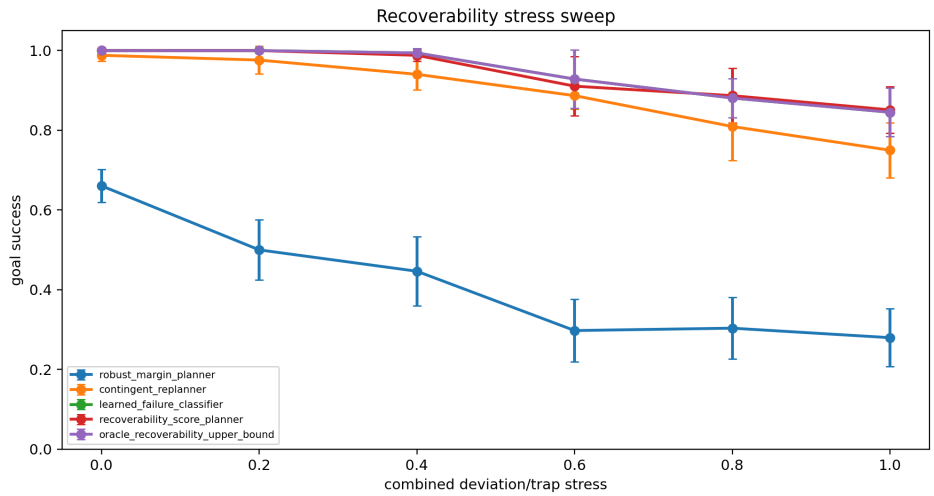

Figure 3: Stress sweeps across five axes. The benchmark is useful for debugging, but the positive claim is not robust enough for ICLR main.

7 Stress Tests

We also sweep navigation difficulty, slip, trap severity, sensor uncertainty, and a combined stress axis. At maximum combined stress, the recoverability score reaches 0.851 goal success, the learned failure classifier reaches 0.845, and the oracle reaches 0.845. This stress slice is favorable to recoverability scoring, but it is not enough to overturn the main split: the proposed method still has nonzero irreversible failures and does not produce a decisive margin.

8 Why This Is Not Submission-Ready

The project fails the ICLR-main readiness gate for three independent reasons.

- **Evidence ceiling.** The benchmark is local and synthetic. It is deterministic and useful, but it is not hardware evidence, not a high-fidelity simulator benchmark, and not an accepted robotics suite.
- **Baseline ceiling.** A simple learned failure classifier is stronger than the proposed method on the main hard split. A positive paper would need a clearer gap against learned predictors and modern TAMP/replanning systems.
- **Objective ceiling.** Ablations show that the proposed objective is misspecified. Removing terms that should matter improves performance.

9 What Would Be Required To Revive It

A future submission-worthy version would need to become a substantially new empirical project:

- evaluate on a real robot or a recognized high-fidelity manipulation/navigation benchmark;
- implement modern task-and-motion planning, contingent planning, and learned world-model baselines;
- train the recoverability score from data rather than hand-designing the score;
- report qualitative rollouts, failure videos, seed-level statistics, paired tests, and calibration curves;
- show that the score reduces irreversible physical errors without hiding cost or damage trade-offs.

10 REPRODUCIBILITY

Run the benchmark from the repository root:

```
python src\run_experiment.py
```

The script writes `rollouts.csv`, `metrics.csv`, `pairwise_stats.csv`, `ablation_metrics.csv`, `stress_sweep.csv`, `negative_cases.csv`, `summary.txt`, and the figures used in this manuscript. The canonical local PDF is `C:/Users/wangz/Downloads/82.pdf`.

11 CONCLUSION

The recoverability idea remains research-interesting: nominal feasibility is plainly inadequate when physical deviations create irreversible traps. But the present v4 evidence does not justify an ICLR-main submission. The correct action is to archive Paper82 as a negative, reproducible evidence audit and move to the next paper.